

Geminos_Feb20

```
library(dagitty)
library(ggdag)
```

Attaching package: 'ggdag'

The following object is masked from 'package:stats':

filter

```
library(ggplot2)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
dag_final <- dagitty("
dag {
  X
  RS
  min_1
  min_2
  min_3

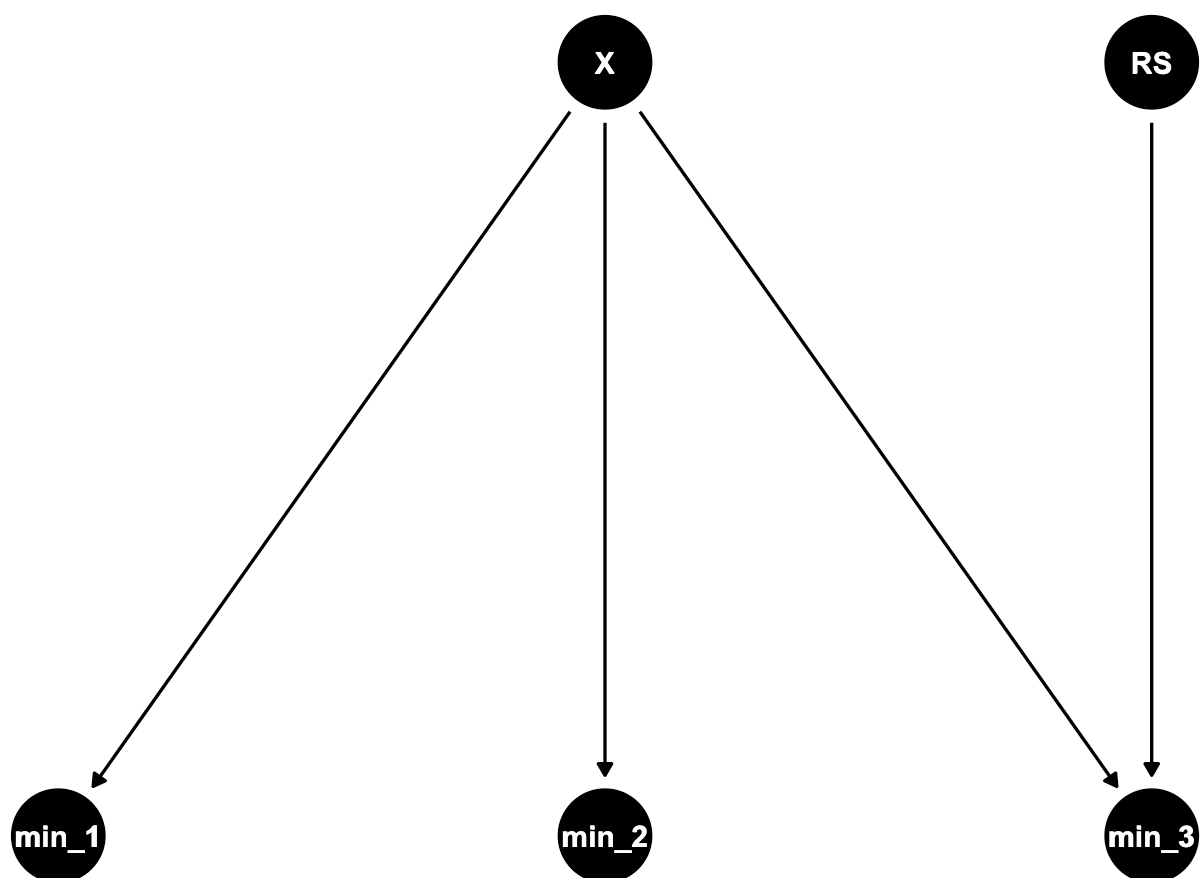
  X -> min_1
  X -> min_2
  X -> min_3

  RS -> min_3
}
")

coordinates(dag_final) <- list(
  x = c(X = 2, RS = 4, min_1 = 0, min_2 = 2, min_3 = 4),
  y = c(X = 3, RS = 3, min_1 = 1, min_2 = 1, min_3 = 1)
)

ggdag(dag_final) +
```

```
theme_dag() +  
geom_dag_text(size = 4)
```



```
set.seed(1)  
n <- 2000  
  
# 1. Covariates (X)  
  
# Age (20–30)  
age <- runif(n, 20, 30)  
  
# Genre preference  
genre_preference <- sample(  
  c("Action", "Drama", "Comedy", "SciFi", "Romance"),  
  n, replace = TRUE,  
  prob = c(0.25, 0.20, 0.30, 0.15, 0.10)  
)  
  
# Major release  
major_release <- sample(  
  c("sports", "politics", "celebrity", "tech", "news"),  
  n, replace = TRUE,  
  prob = c(0.30, 0.20, 0.30, 0.10, 0.10)
```

```
)

# -----
# 2. Treatment depends on X
# -----

genre_effect <- c(
  Action = 0.4,
  Drama = -0.3,
  Comedy = 0.2,
  SciFi = 0.5,
  Romance = -0.2
)

release_effect <- c(
  sports = 0.3,
  politics = -0.4,
  celebrity = 0.6,
  tech = 0.2,
  news = -0.1
)

linpred <-
  0.08*(age - 25) +
  genre_effect[genre_preference] +
  release_effect[major_release]

p <- plogis(linpred)

RS <- rbinom(n, 1, p)

# 3. Baseline outcome (Y1 = min_2)

min_2 <-
  0.4*(age - 25) +
  0.5*genre_effect[genre_preference] +
  0.3*release_effect[major_release] +
  rnorm(n)

# 4. Post outcome (Y2 = min_3)

min_3 <-
  1 +
  0.6*(age - 25) +
  0.6*genre_effect[genre_preference] +
  0.4*release_effect[major_release] +
  2*RS +
  rnorm(n)

# 5. Final dataset
```

```
data <- data.frame(
  age,
  genre_preference,
  major_release,
  RS,
  min_2,
  min_3
)

data$diff <- data$min_3 - data$min_2
```

1. Theorem 2.1, unconditional DiD

```
ETT_1 <-
  (mean(data$min_3[data$RS==1]) -
   mean(data$min_2[data$RS==1])) -
  (mean(data$min_3[data$RS==0]) -
   mean(data$min_2[data$RS==0]))
```

```
ETT_1
```

```
[1] 2.241722
```

2. Theorem 2.3, regression adjustment

```
term1 <- mean(data$diff[data$RS==1])

control_model <- lm(
  diff ~ age + genre_preference + major_release,
  data = subset(data, RS==0)
)

m0_pred <- predict(
  control_model,
  newdata = subset(data, RS==1)
)

ETT_2 <- term1 - mean(m0_pred)
ETT_2
```

```
[1] 2.093596
```

3. Theorem 2.4

```
ps_model <- glm(
  RS ~ age + genre_preference + major_release,
```

```
data = data,  
family = binomial  
)  
  
data$ps <- predict(ps_model, type="response")  
  
ETT_3 <- mean(  
  (data$diff / mean(data$RS)) *  
  (data$RS - data$ps) /  
  (1 - data$ps)  
)  
  
ETT_3
```

[1] 2.096087

```
c(  
  "2.1_DiD" = ETT_1,  
  "2.3_RegAdj" = ETT_2,  
  "2.4_IPW" = ETT_3  
)
```

2.1_DiD	2.3_RegAdj	2.4_IPW
2.241722	2.093596	2.096087